

Week 4: Differentiability, Tangent Planes, and the Gradient

Differentiability and Tangent Planes

- Find the equation of the tangent line at the given points:
 - $f(x, y) = x^2 + y^{-2}$, $(4, 1)$
 - $F(r, s) = r^2 s^{-1/2} + s^{-3}$, $(2, 1)$
- Find the points on the graph of $f(x, y) = 3x^2 - xy - y^2$ at which the tangent plane is horizontal
- Find the linearization of $f(x, y, z) = z\sqrt{x + y}$ centered at $(8, 4, 5)$.
- Use the linear approximation to estimate $\sqrt{(1.9)(2.02)(4.05)}$
- The monthly payment for a home loan is given by a function $f(P, r, N)$ where P is the principal (initial size of the loan), r the interest rate, and N the length of the loan in months. Interest rates are expressed as a decimal: A 6% interest rate is denoted by $r=0.06$. If $P = \$100,000$, $r= 0.06$, and $N = 240$ (a 20-year loan), then the monthly payment is $f(100000, 0.06, 240) = 716.43$. Furthermore, at these values, we have
$$\frac{\delta f}{\delta P} = 0.0071, \frac{\delta f}{\delta r} = 5769, \frac{\delta f}{\delta N} = -1.5467$$
Estimate:
 - The change in monthly payment per \$1000 increase in loan principal
 - The change in monthly payment if the interest rate increases to $r = 6.5\%$ and $r = 7\%$
 - The change in monthly payment if the length of the loan increases to 24 years

The Gradient and Directional Derivatives

1. The temperature at a point (x, y, z) in 3-space is given in degrees Celsius by $T(x, y, z) = e^{-(x^2+y^2+z^2)}$.
 - a. Describe in words the shape of the surfaces on which the temperature is constant.
 - b. Find ∇T .
 - c. A particle travels from the point $(1, 0, 0)$ to the point $(2, 1, 0)$ at a speed of 3 units per second. Find the instantaneous rate of change of the temperature as you leave the point $(1, 0, 0)$, including units.
2. Given three functions f , g , and h below, find:
 - a. $f(x, y, z) = x^2 + 2xyz + y^2z$
 - b. $g(x, y, z) = \frac{xy}{z}$
 - c. $h(x, y, z) = \ln(x^2 + y^2 + z^2)$
 - a. ∇f
 - b. ∇g
 - c. $\nabla(f + g)$
 - d. $\nabla(fg)$
 - e. ∇h
3. Find the two points on the ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$ where the tangent plane is normal to $\langle 1, 1, -2 \rangle$.

4. Marvin the mayfly is flying around the S unbearably hot. In fact, he estimates that $x^3 + 2y^{-1} + 4\ln(z)$ and he is currently at po that he can cool off?